Neck vessels & nerves

Carotid and jugular systems, the subclavian, the last four cranial nerves, cervical plexus, phrenic nerve and sympathetic chain.

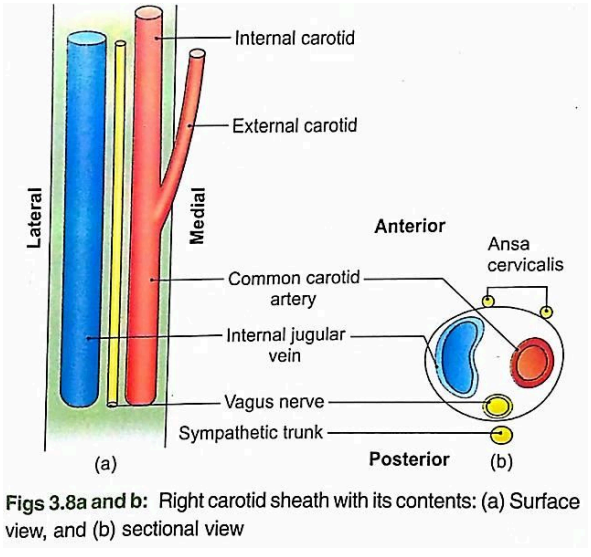
1. Carotid sheath and common carotid
2. External carotid artery
3. Internal carotid artery
4. Veins of the neck
5. Root of the neck: subclavian artery and thoracic duct
6. Glossopharyngeal nerve (IX)
7. Vagus nerve (X)
8. Accessory nerve (XI) and hypoglossal nerve (XII)
9. Cervical plexus, ansa cervicalis and phrenic nerve
10. Cervical sympathetic chain
11. Superior thyroid and lingual arteries
12. Facial artery and facial vein
13. Occipital, posterior auricular, ascending pharyngeal and superficial temporal arteries
14. Vertebral artery and the subclavian branches
15. External and anterior jugular veins
16. Horner's syndrome: the oculosympathetic pathway
17. Variations surgeons meet
18. In theatre
19. On imaging

01 Carotid sheath and common carotid

- Sheath contains the **common/internal carotid medially, IJV laterally, vagus between and behind them**; the ansa cervicalis lies on or in its front; the sympathetic chain is behind it.
- **Common carotid:** right from the brachiocephalic trunk, left from the arch. Ascends to the **upper border of the thyroid cartilage (C3–C4)**, where it divides. No branches in the neck.
- **Carotid sinus** (baroreceptor) at the ICA origin and **carotid body** (chemoreceptor) in the bifurcation, both supplied by **IX** (nerve of Hering).

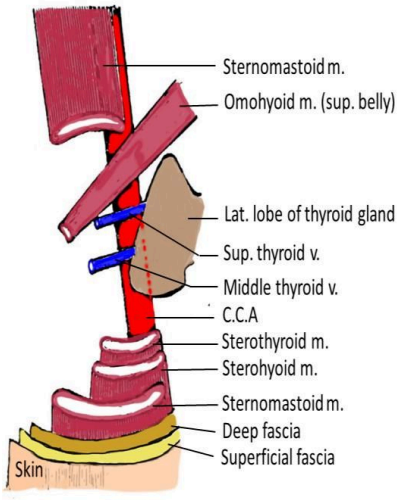
CLINICAL PEARL Carotid sinus massage slows the heart; a carotid body tumour splays the ICA and ECA apart (lyre sign).

SOURCES DIFFER ORDER IN THE SHEATH: Doss: CCA anteromedial, IJV posterolateral, vagus between and behind (as here). ENT Steps lists superficial to deep as IJV, vagus, CCA.

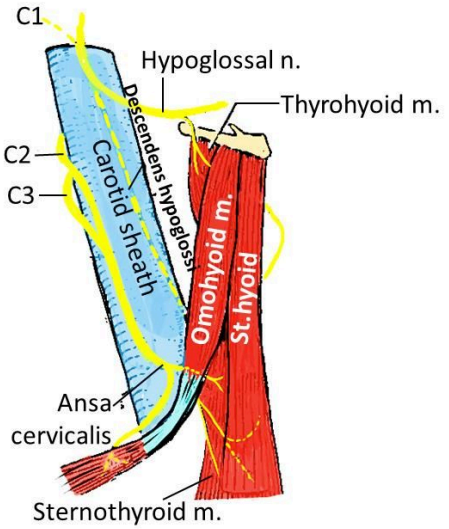


Figs 3.8a and b: Right carotid sheath with its contents: (a) Surface view, and (b) sectional view

Carotid sheath and its contents
ENT Steps Anatomy, p. 266

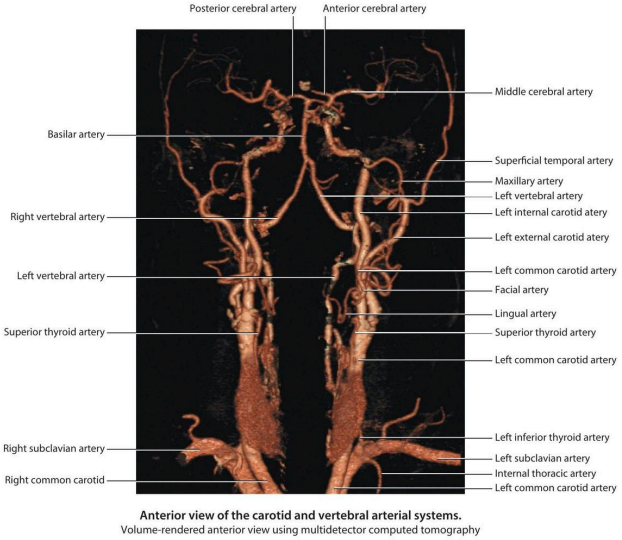


Relations of the common carotid artery
Dr. Sameh Doss handout, p. 136



Ansa cervicalis and the carotid sheath

Dr. Sameh Doss handout, p. 124



Carotid and vertebral arteries: CT angiography, anterior view

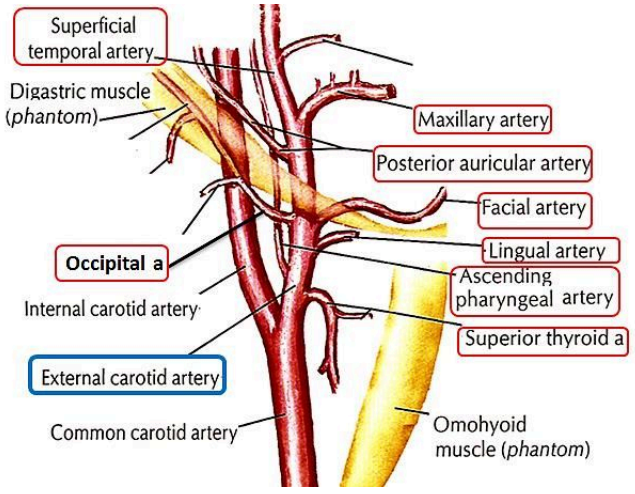
Gray's Atlas of Anatomy, 2nd ed., p. 501

02 External carotid artery

- Begins in front of and medial to the ICA; ends in the parotid behind the neck of the mandible by dividing into the **maxillary** and **superficial temporal** arteries.
- **Branches ("Some Anatomists Like Freaking Out Poor Medical Students"):** Superior thyroid, Ascending pharyngeal, Lingual, Facial, Occipital, Posterior auricular, Maxillary, Superficial temporal.
- The ECA has branches in the neck; the ICA does not. This is how you tell them apart before ligation.
- **Lingual artery:** loops above the greater horn of the hyoid and runs **deep to hyoglossus**, while the hypoglossal nerve runs **on** the hyoglossus.
- **Facial artery:** grooves the submandibular gland, then crosses the mandible at the front of the masseter.

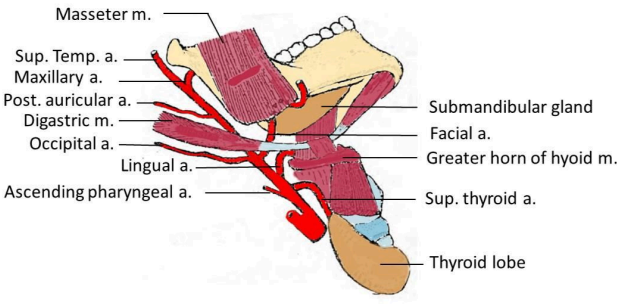
CLINICAL PEARL Hypoglossal nerve crosses superficial to both carotids, looping round the occipital artery's SCM branch: protect it when ligating the ECA.

SOURCES DIFFER FIRST BRANCH: ENT Steps and Gray's say the superior thyroid; Doss calls the ascending pharyngeal the first branch (from the medial side).



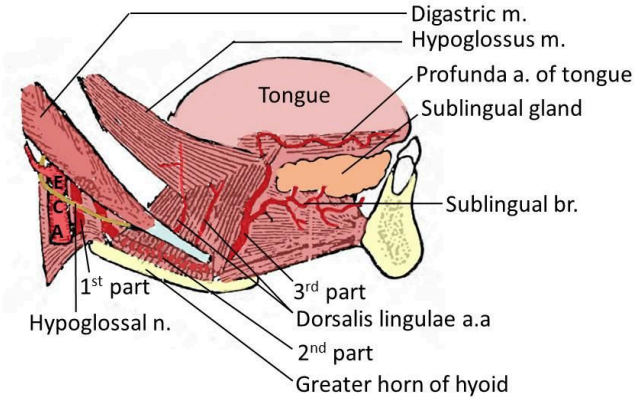
Branches of the external carotid artery

ENT Steps Anatomy, p. 267



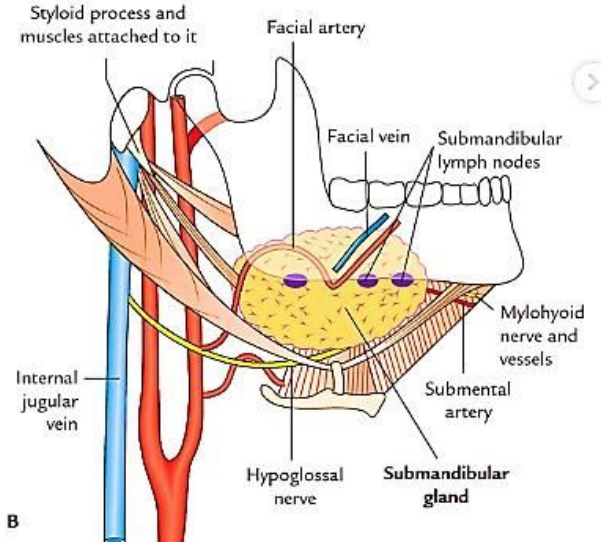
Branches of the external carotid artery

Dr. Sameh Doss handout, p. 142



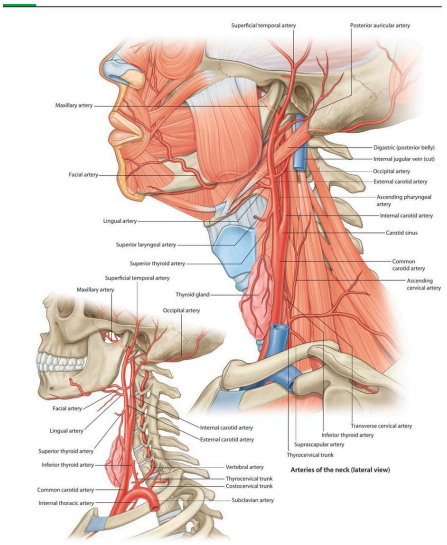
Lingual artery and hypoglossal nerve

Dr. Sameh Doss handout, p. 142



Submandibular triangle: facial vessels and hypoglossal nerve

ENT Steps Anatomy, p. 271



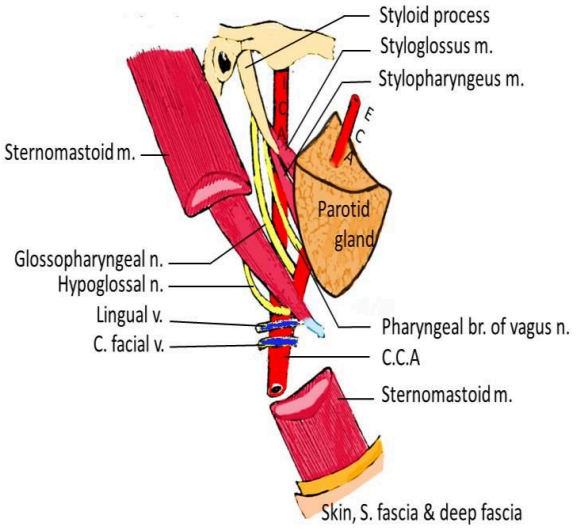
Arteries of the neck (lateral view)

Gray's Atlas of Anatomy, 2nd ed., p. 546

03 Internal carotid artery

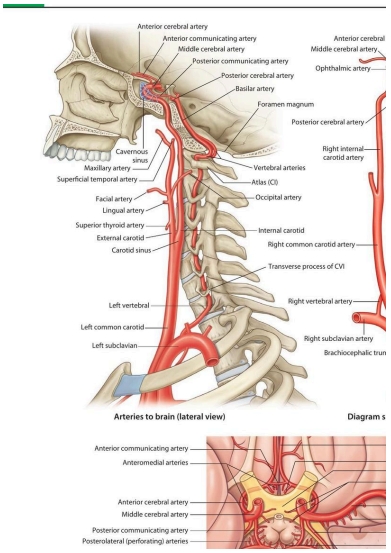
- Cervical part runs up in the carotid sheath to the carotid canal, **without branches**. Then petrous, lacerum, cavernous and cerebral parts.
- **Behind it:** longus capitis, superior cervical ganglion, superior laryngeal nerve. **Between ICA and ECA:** styloglossus, stylopharyngeus, IX and the pharyngeal branch of X.
- Lies about 2.5 cm behind and lateral to the tonsil; aberrant medial loops can lie under the posterior pharyngeal wall.

> **CLINICAL PEARL** A pulsatile tonsil or posterior pharyngeal wall can be an aberrant ICA: do not incise it.



Internal carotid artery in the neck

Dr. Sameh Doss handout, p. 138



Arteries to the brain (lateral view): carotid and vertebral systems

Gray's Atlas of Anatomy, 2nd ed., p. 500

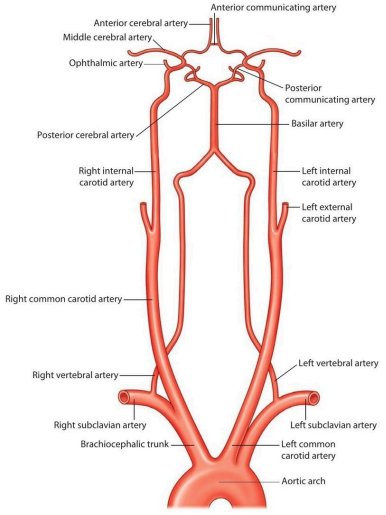
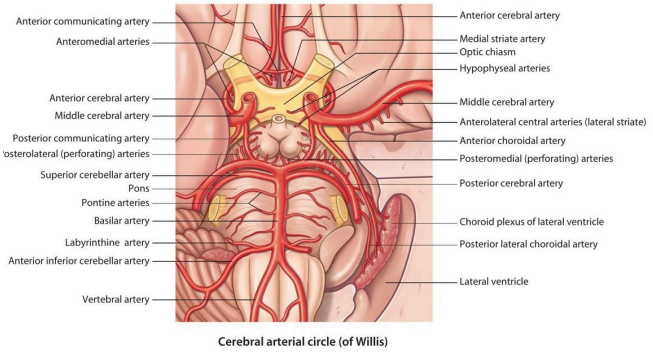


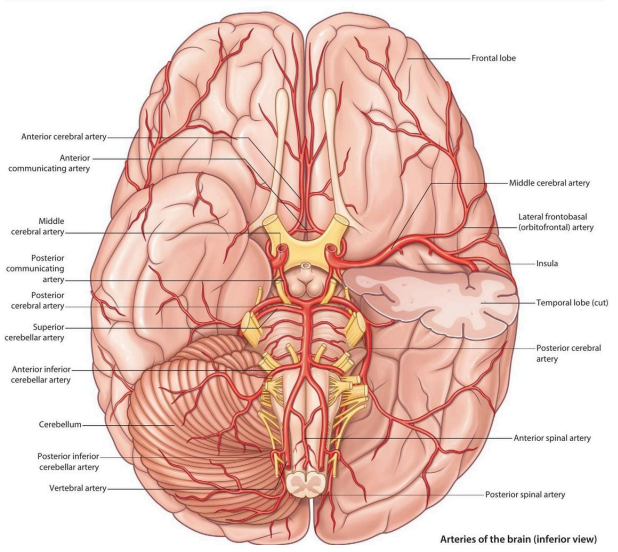
Diagram of the blood supply to the brain: carotid and vertebral systems

Gray's Atlas of Anatomy, 2nd ed., p. 500



Cerebral arterial circle (of Willis)

Gray's Atlas of Anatomy, 2nd ed., p. 500

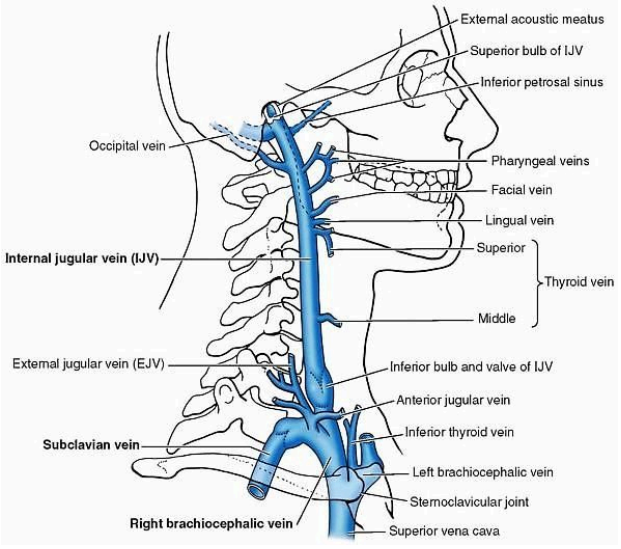


Arteries of the brain (inferior view): vertebral, basilar and their branches

Gray's Atlas of Anatomy, 2nd ed., p. 502

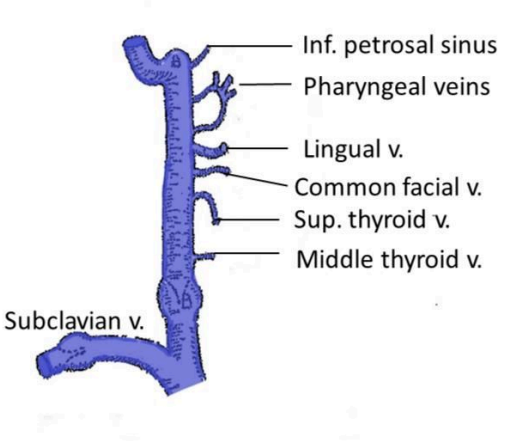
04 Veins of the neck

- **IJV:** continuation of the **sigmoid sinus** at the jugular foramen; runs in the carotid sheath to join the subclavian vein behind the sternoclavicular joint (forming the brachiocephalic vein).
- **Tributaries:** inferior petrosal sinus, pharyngeal, common facial, lingual, superior and middle thyroid veins.
- **External jugular vein:** from the posterior division of the retromandibular vein and the posterior auricular vein; crosses SCM to drain into the subclavian vein.
- **Anterior jugular veins** join across the suprasternal space (jugular arch).



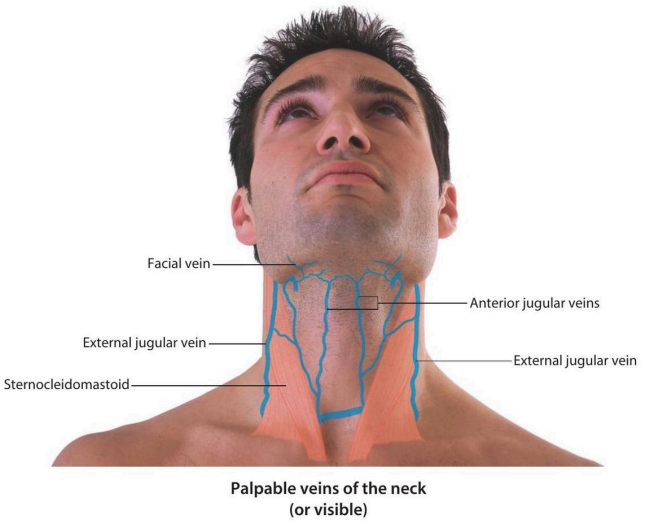
Veins of the head and neck

ENT Steps Anatomy, p. 268



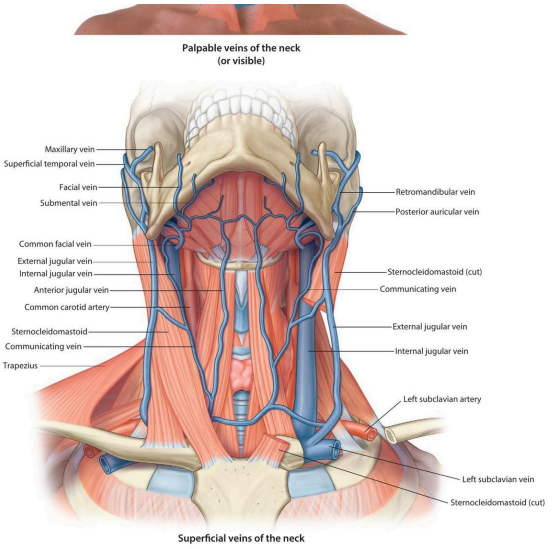
Tributaries of the internal jugular vein

Dr. Sameh Doss handout, p. 146



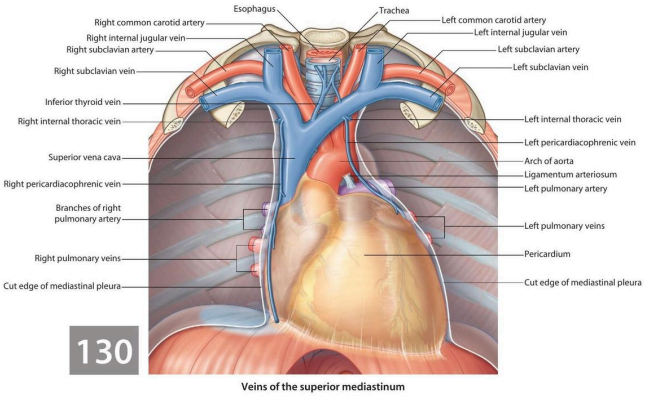
Palpable veins of the neck

Gray's Atlas of Anatomy, 2nd ed., p. 539



Superficial veins of the neck

Gray's Atlas of Anatomy, 2nd ed., p. 539



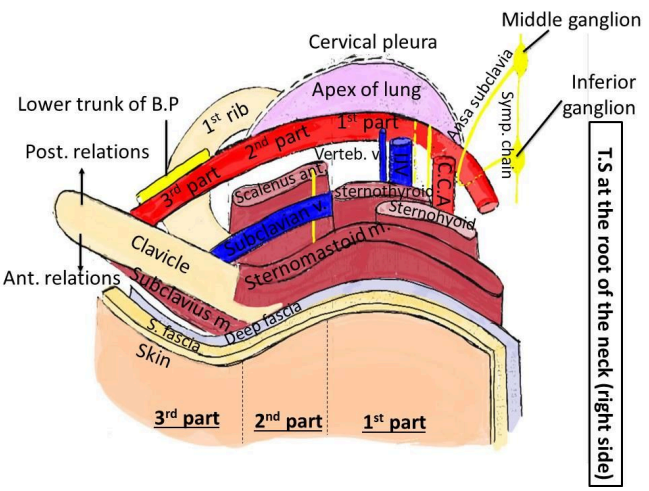
Veins of the superior mediastinum: brachiocephalic veins and their tributaries

Gray's Atlas of Anatomy, 2nd ed., p. 109

05 Root of the neck: subclavian artery and thoracic duct

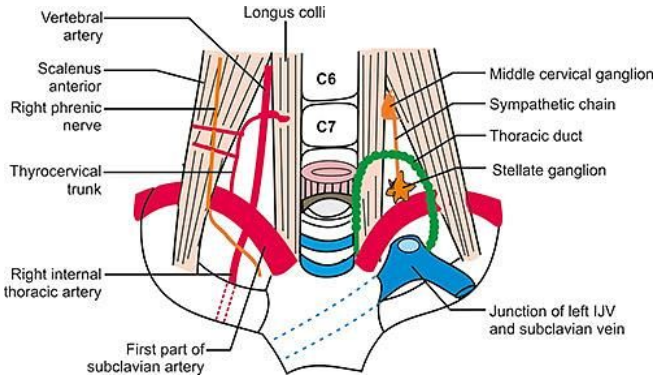
- **Subclavian artery:** first part (medial to scalenus anterior) gives the **vertebral**, **internal thoracic** and **thyrocervical trunk** (inferior thyroid, transverse cervical, suprascapular); second part (behind the muscle) gives the costocervical trunk; third part (lateral) usually none.
- **Inferior thyroid artery** arches medially behind the carotid sheath to the back of the thyroid, crossing the RLN.
- **Thoracic duct** arches on the left at C7 to enter the junction of the IJV and subclavian vein.
- **Chassaignac's (scalenovertebral) triangle:** longus colli, scalenus anterior and subclavian artery, apex at the C6 tubercle.

> **CLINICAL PEARL** Milky drainage after a left neck dissection means a chyle leak from the thoracic duct.



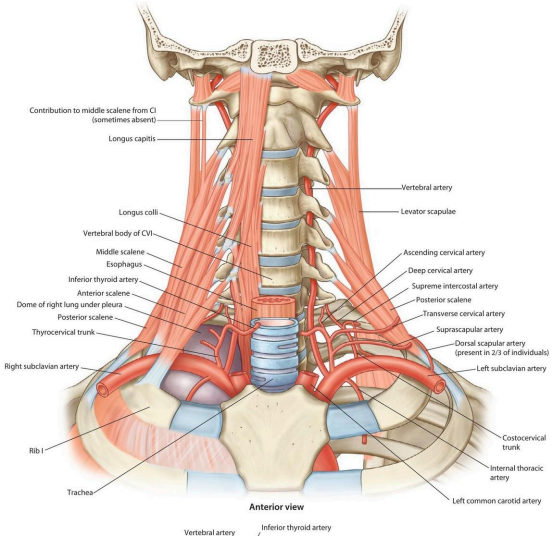
Subclavian artery and its three parts

Dr. Sameh Doss handout, p. 131



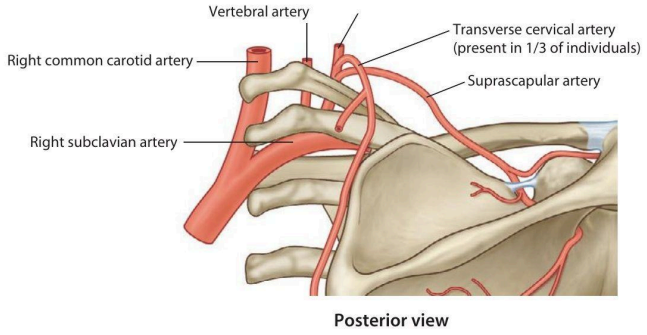
Root of the neck: subclavian artery, thoracic duct, phrenic nerve

ENT Steps Anatomy, p. 269



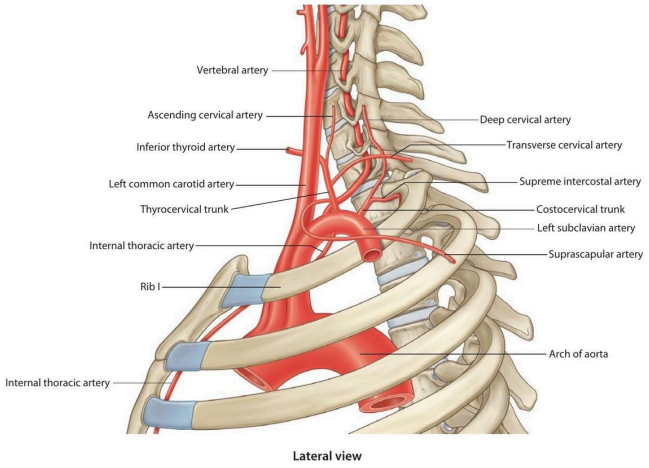
Root of the neck: arteries (anterior view)

Gray's Atlas of Anatomy, 2nd ed., p. 548



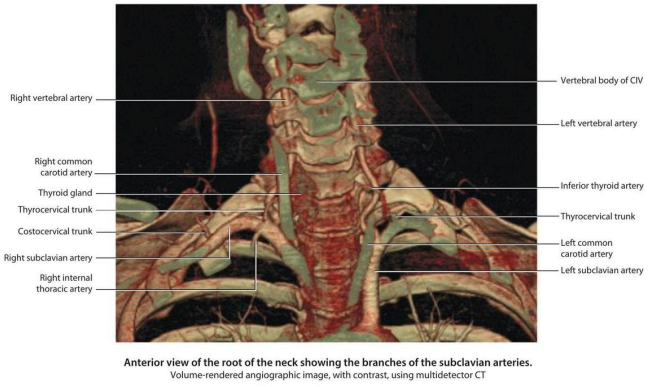
Root of the neck: thyrocervical trunk (posterior view)

Gray's Atlas of Anatomy, 2nd ed., p. 548



Root of the neck: arteries (lateral view)

Gray's Atlas of Anatomy, 2nd ed., p. 549



Anterior view of the root of the neck showing the branches of the subclavian arteries.
Volume-rendered angiographic image, with contrast, using multidetector CT

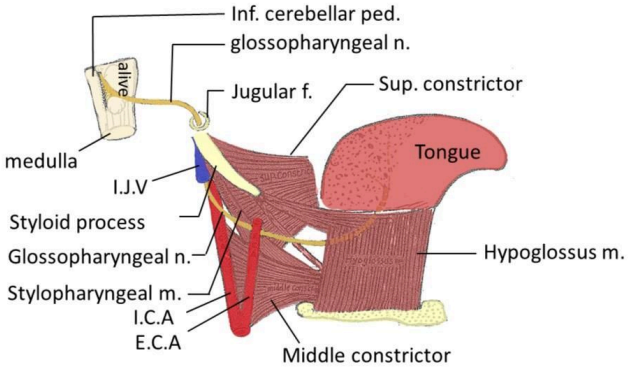
Branches of the subclavian arteries: CT angiography

Gray's Atlas of Anatomy, 2nd ed., p. 549

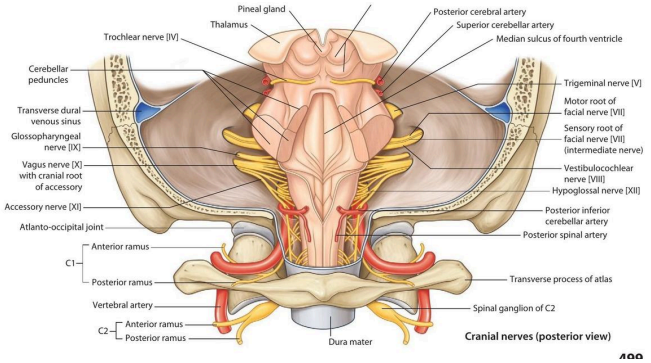
06 Glossopharyngeal nerve (IX)

- Leaves through the jugular foramen; curves forward between the ICA and ECA on **stylopharyngeus** (its only muscle) to the tongue base.
- **Branches:** tympanic (Jacobson's) → lesser petrosal → parotid; carotid sinus nerve; pharyngeal (sensory to the plexus); tonsillar; lingual (taste and sensation, posterior third).
- Afferent limb of the **gag reflex**.

> **CLINICAL PEARL** Glossopharyngeal neuralgia: lancinating pain in the tonsil and ear on swallowing.



Glossopharyngeal nerve
Dr. Sameh Doss handout, p. 151

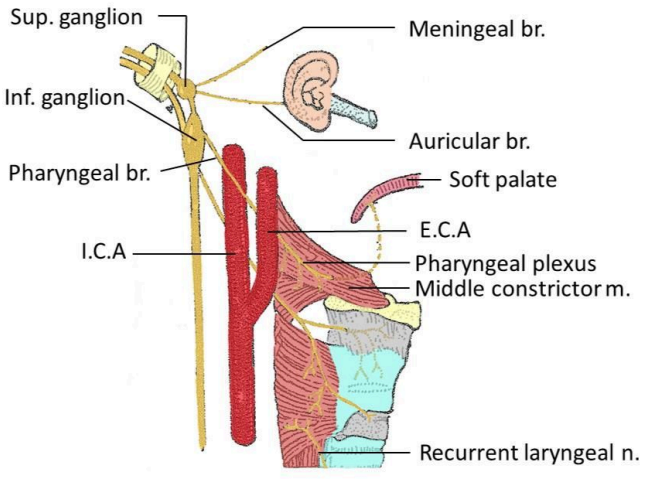


Cranial nerves (posterior view): IX, X, XI and XII at the jugular foramen
Gray's Atlas of Anatomy, 2nd ed., p. 499

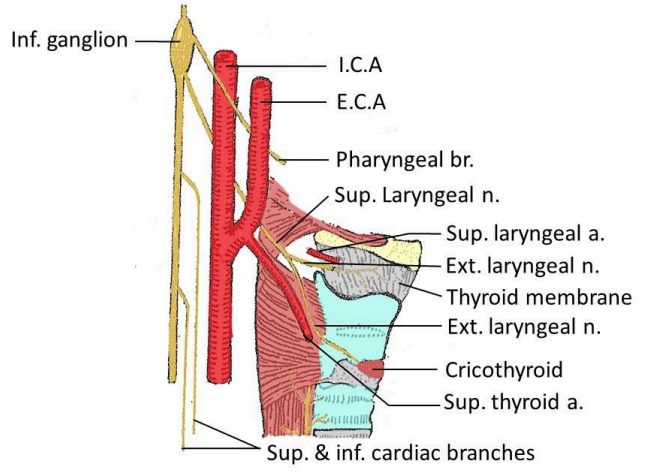
07 Vagus nerve (X)

- Leaves the jugular foramen with **superior and inferior ganglia**; descends in the carotid sheath behind and between the artery and vein.
- **Branches in the neck:** auricular (Arnold's), pharyngeal (motor to the pharyngeal plexus, soft palate except tensor), **superior laryngeal** (internal and external), cardiac branches, **recurrent laryngeal**.
- Right RLN hooks under the subclavian artery; left under the arch of the aorta.

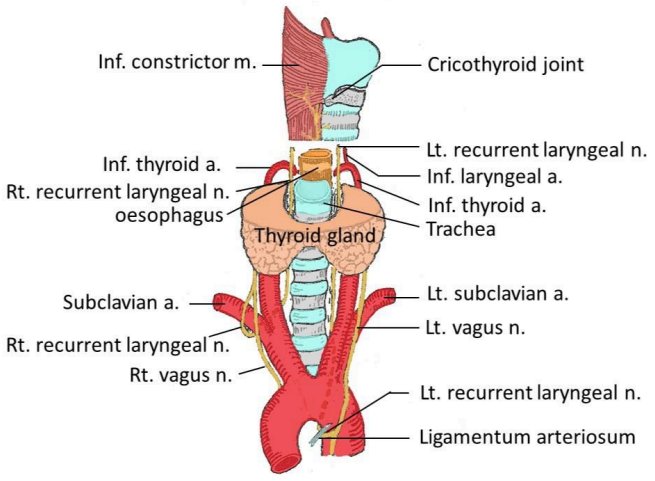
> **CLINICAL PEARL** A high vagal lesion (above the pharyngeal branch) also paralyses the palate: the uvula deviates to the healthy side.



Vagus nerve in the neck
Dr. Sameh Doss handout, p. 153

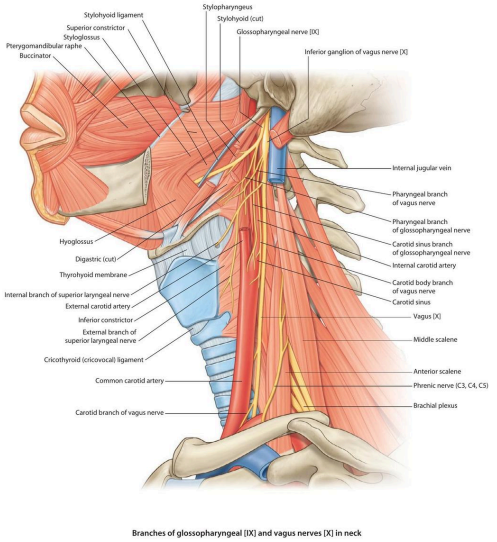


Branches of the vagus nerve in the neck
Dr. Sameh Doss handout, p. 154



Recurrent laryngeal nerves and inferior thyroid arteries

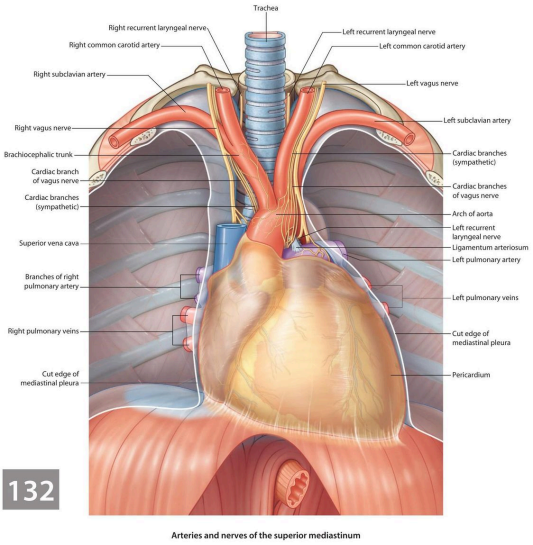
Dr. Sameh Doss handout, p. 154



Branches of glossopharyngeal [IX] and vagus nerves [X] in neck

Branches of the glossopharyngeal [IX] and vagus [X] nerves in the neck

Gray's Atlas of Anatomy, 2nd ed., p. 543



Arteries and nerves of the superior mediastinum

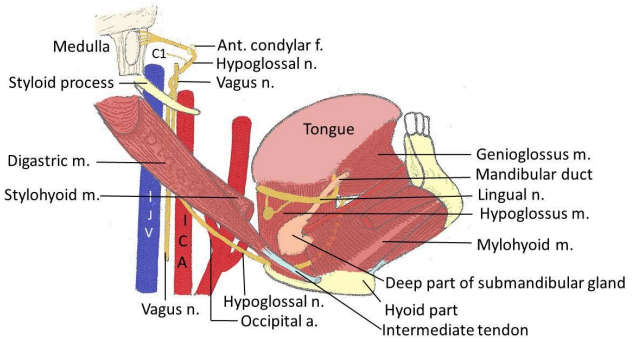
Arteries and nerves of the superior mediastinum: vagus and recurrent laryngeal loops

Gray's Atlas of Anatomy, 2nd ed., p. 110

08 Accessory nerve (XI) and hypoglossal nerve (XII)

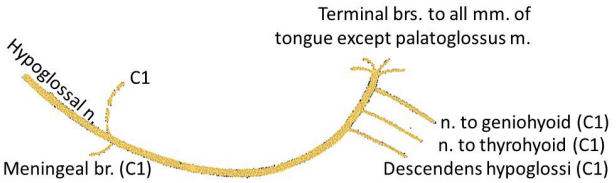
- **XI (spinal part):** from C1–C5 up through the foramen magnum and out of the jugular foramen; runs back across (usually in front of) the IJV, deep to the posterior belly of digastric, into **SCM**, then across the posterior triangle to **trapezius**.
- **XII:** leaves the hypoglossal canal, passes between ICA and IJV, loops round the occipital artery, crosses **superficial to the ICA and ECA** above the bifurcation, runs on hyoglossus below the lingual nerve and Wharton's duct.
- C1 fibres travel with XII and leave as the **superior root of the ansa cervicalis**, the nerve to thyrohyoid and geniohyoid.

> **CLINICAL PEARL** In level IIA dissection the accessory nerve is found where it crosses the IJV, just below the posterior belly of digastric.



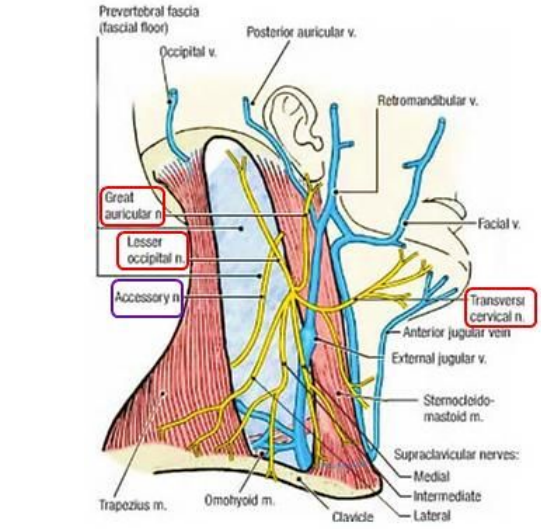
Hypoglossal nerve

Dr. Sameh Doss handout, p. 156



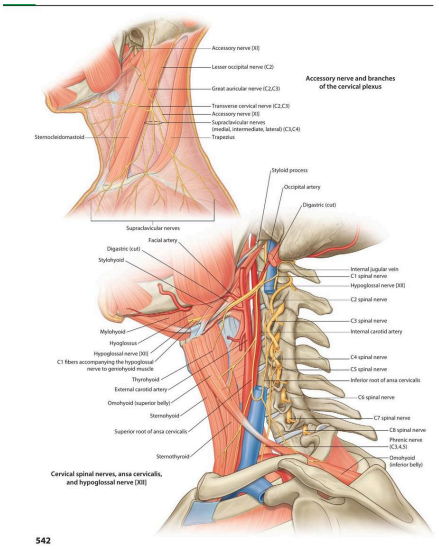
Branches of the hypoglossal nerve

Dr. Sameh Doss handout, p. 157



Nerves of the neck: cervical plexus and accessory nerve

ENT Steps Anatomy, p. 265



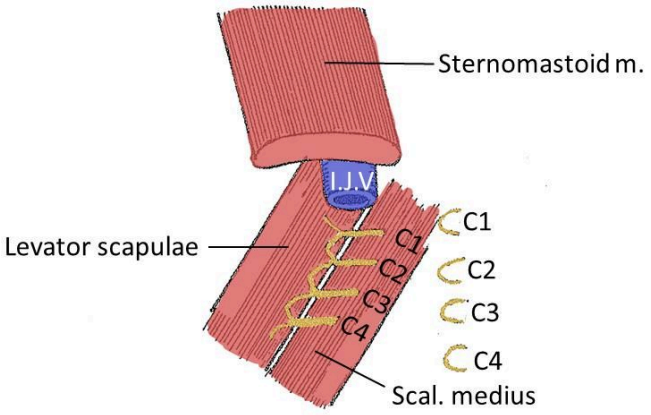
Accessory nerve, cervical plexus, ansa cervicalis and hypoglossal nerve

Gray's Atlas of Anatomy, 2nd ed., p. 542

09 Cervical plexus, ansa cervicalis and phrenic nerve

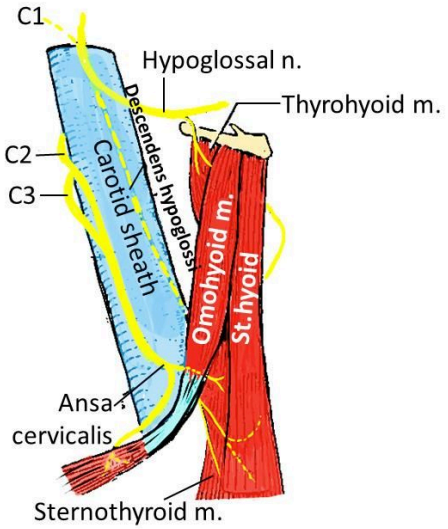
- **Cervical plexus (C1–C4):** cutaneous branches at Erb's point: **lesser occipital (C2), great auricular (C2–3), transverse cervical (C2–3), supraclavicular (C3–4)**.
- **Ansa cervicalis:** superior root (C1 via XII) + inferior root (C2–3); supplies the strap muscles except thyrohyoid.
- **Phrenic nerve (C3–C5):** runs down on **scalenus anterior** deep to the prevertebral fascia, behind the transverse cervical and suprascapular arteries.

> **CLINICAL PEARL** "C3, 4, 5 keeps the diaphragm alive." Staying above the prevertebral fascia protects the phrenic nerve and brachial plexus in neck dissection.



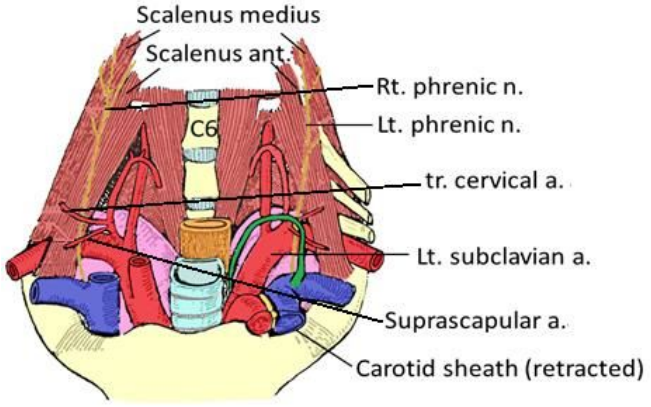
Cervical plexus

Dr. Sameh Doss handout, p. 158



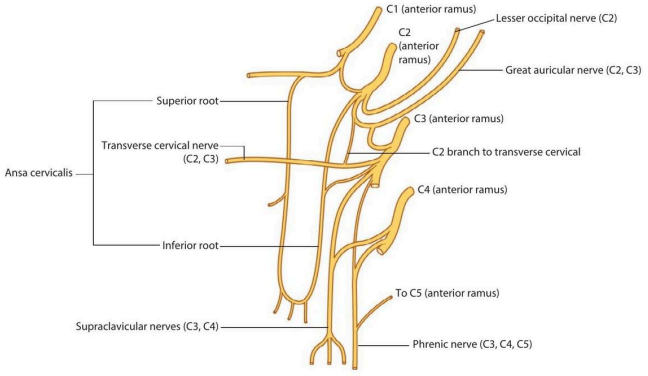
Ansa cervicalis and the carotid sheath

Dr. Sameh Doss handout, p. 124



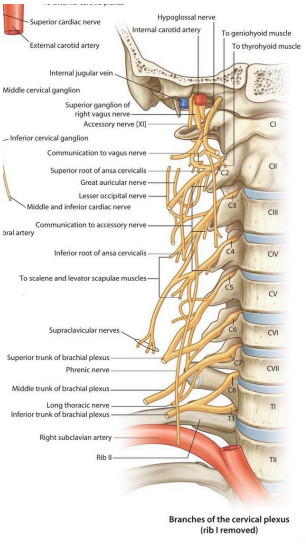
Phrenic nerves on scalenus anterior

Dr. Sameh Doss handout, p. 159



Cervical plexus

Gray's Atlas of Anatomy, 2nd ed., p. 544



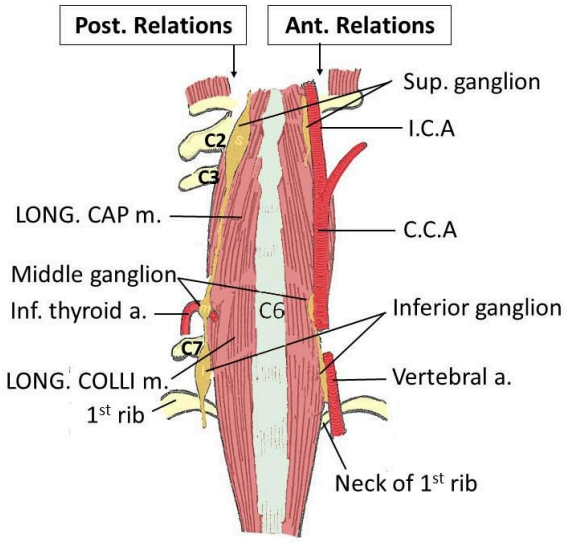
Branches of the cervical plexus (rib I removed)

Gray's Atlas of Anatomy, 2nd ed., p. 545

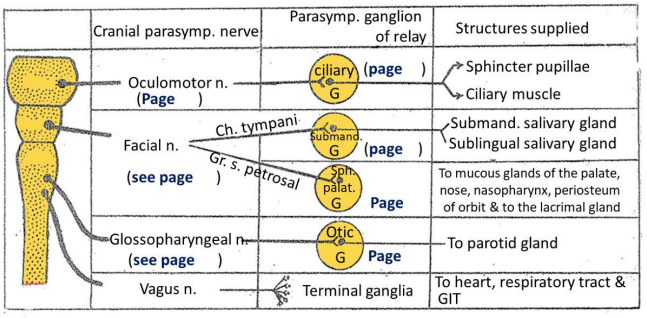
10 Cervical sympathetic chain

- Lies behind the carotid sheath on longus colli/capitis. **Superior cervical ganglion** (C2–3, largest), **middle** (C6), **inferior** (often fused with T1 as the **stellate ganglion**).
- Postganglionic fibres follow the ICA and ECA to the head: pupil dilator, tarsal muscle, sweat glands, blood vessels, salivary glands.
- Cranial parasympathetic ganglia: ciliary (III), **pterygopalatine (VII)**, **submandibular (VII)**, **otic (IX)**.

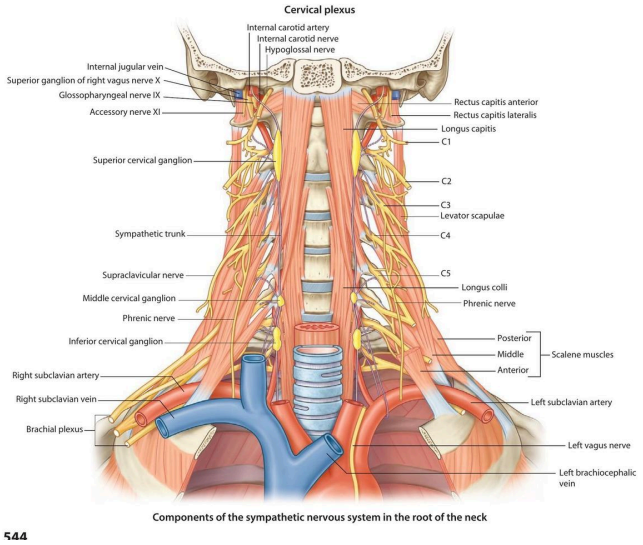
CLINICAL PEARL Horner's syndrome (ptosis, miosis, anhidrosis, enophthalmos) can follow neck surgery or a mass compressing the chain.



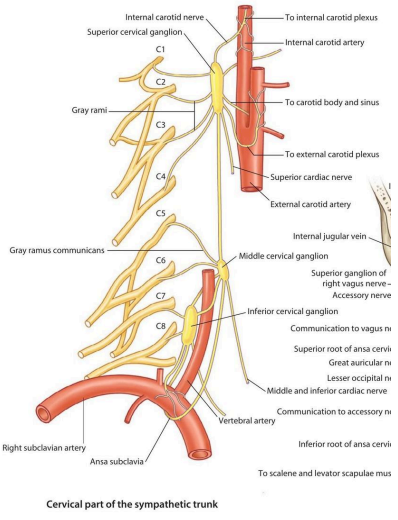
Cervical sympathetic chain
Dr. Sameh Doss handout, p. 160



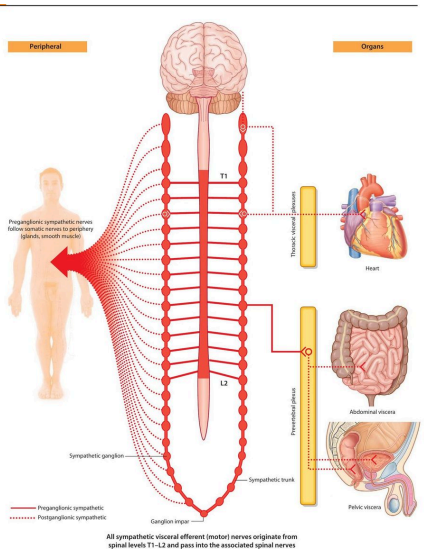
Cranial parasympathetic ganglia
Dr. Sameh Doss handout, p. 159



Sympathetic system in the root of the neck
Gray's Atlas of Anatomy, 2nd ed., p. 544



Cervical part of the sympathetic trunk
Gray's Atlas of Anatomy, 2nd ed., p. 545



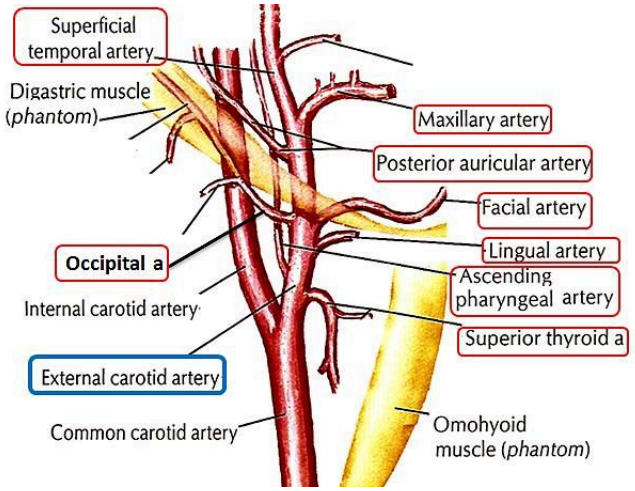
Sympathetic outflow: T1 to L2, the sympathetic trunk and its targets

Gray's Atlas of Anatomy, 2nd ed., p. 14

11 Superior thyroid and lingual arteries

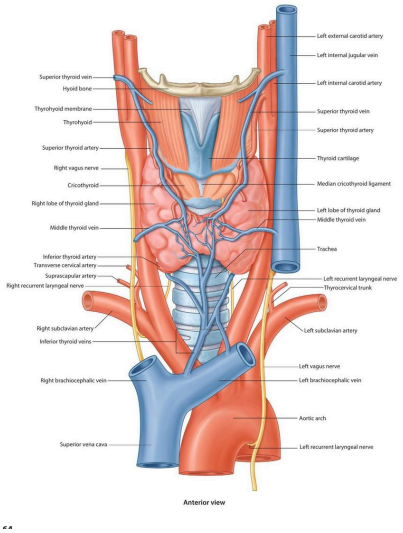
- **Superior thyroid artery:** usually the **first branch** of the external carotid (sometimes from the bifurcation or the common carotid), arising just below the greater horn of the hyoid. It runs down and forward under the infrahyoid strap muscles to the **upper pole**.
- **Its branches:** infrahyoid, sternocleidomastoid, the **superior laryngeal artery** (pierces the thyrohyoid membrane with the internal laryngeal nerve), a **cricothyroid** branch across the upper cricothyroid membrane, and anterior and posterior glandular branches.
- **External laryngeal nerve:** runs with the artery on the inferior constrictor towards cricothyroid. It may cross between the artery's branches close to the upper pole (Cernea types 2a and 2b), so the branches are ligated individually on the capsule.
- **Lingual artery:** arises opposite the tip of the greater horn of the hyoid and makes an upward **loop**, crossed superficially by the hypoglossal nerve.
- It then runs **deep to hyoglossus**, while the hypoglossal nerve, submandibular duct and lingual nerve lie **on** the muscle. It can be found here in **Lesser's triangle** (the two bellies of digastric and the hypoglossal nerve).
- **Lingual branches:** suprahyoid, **dorsal lingual** (posterior tongue, tonsil, soft palate), **sublingual** (gland and floor of mouth), and the terminal **deep lingual (ranine) artery**, which runs to the tip beside the frenulum.

> **CLINICAL PEARL** Mass ligation of the upper pole risks the external laryngeal nerve: the voice loses its high notes and tires easily, often with a normal-looking larynx.



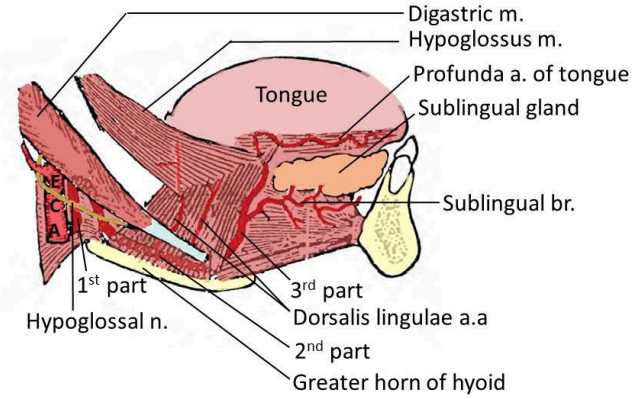
Branches of the external carotid artery

ENT Steps Anatomy, p. 267



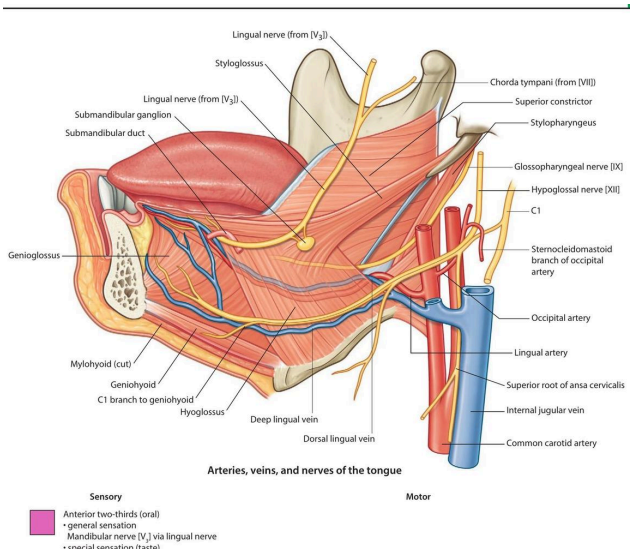
Vasculature of the thyroid gland (anterior view)

Gray's Atlas of Anatomy, 2nd ed., p. 564



Lingual artery and hypoglossal nerve

Dr. Sameh Doss handout, p. 142



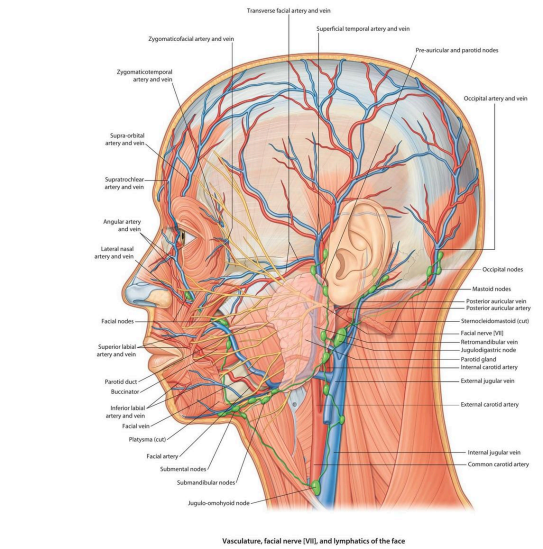
Arteries, veins and nerves of the tongue

Gray's Atlas of Anatomy, 2nd ed., p. 579

12 Facial artery and facial vein

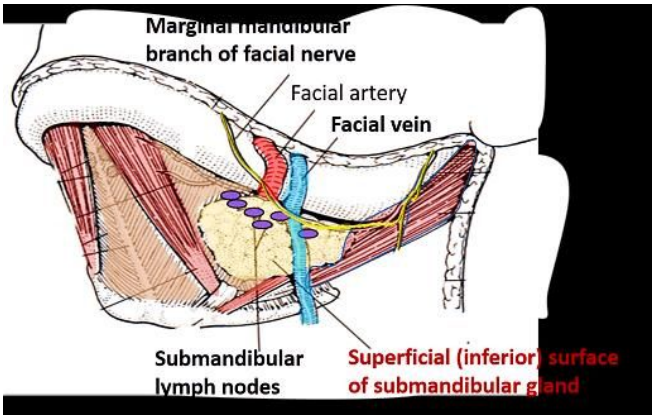
- **Facial artery:** tortuous, it loops past the tonsil (tonsillar and ascending palatine branches), grooves the deep surface of the submandibular gland, hooks over the mandible at the front of masseter, and ends as the **angular artery** at the medial canthus.
- **Facial vein:** begins as the **angular vein** (supratrochlear plus supra-orbital veins). It runs a straighter course **behind** the artery and crosses the mandible with it.
- In the neck the vein passes **superficial** to the submandibular gland (the artery is deep to it). It joins the anterior division of the retromandibular vein to form the **common facial vein**, which drains into the IJV near the hyoid.
- **Links to the cavernous sinus:** the angular vein communicates with the **superior ophthalmic vein**, and the **deep facial vein** joins the pterygoid plexus. Flow in these veins can reverse, so infection of the upper lip and nose (the danger area) can spread to the sinus.
- **Marginal mandibular nerve:** crosses **superficial** to the facial artery and vein at the lower border of the mandible. Dividing the vein low and turning it up with the flap (the **Hayes Martin manoeuvre**) lifts the nerve out of harm's way in submandibular surgery.

CLINICAL PEARL Infection in the danger area can spread along the angular and superior ophthalmic veins to the cavernous sinus: watch for proptosis, chemosis, ophthalmoplegia (VI first) and a swinging fever.



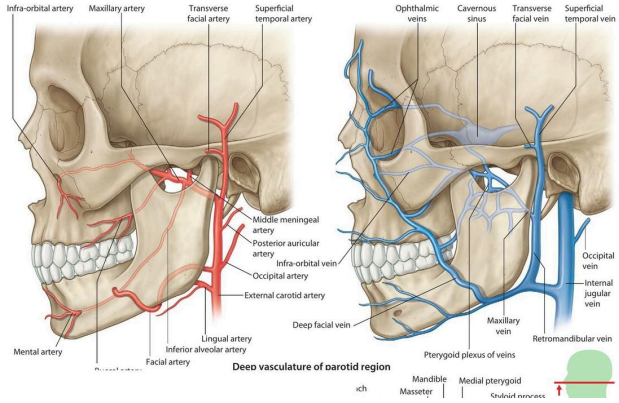
Vasculature, facial nerve [VII] and lymphatics of the face

Gray's Atlas of Anatomy, 2nd ed., p. 506



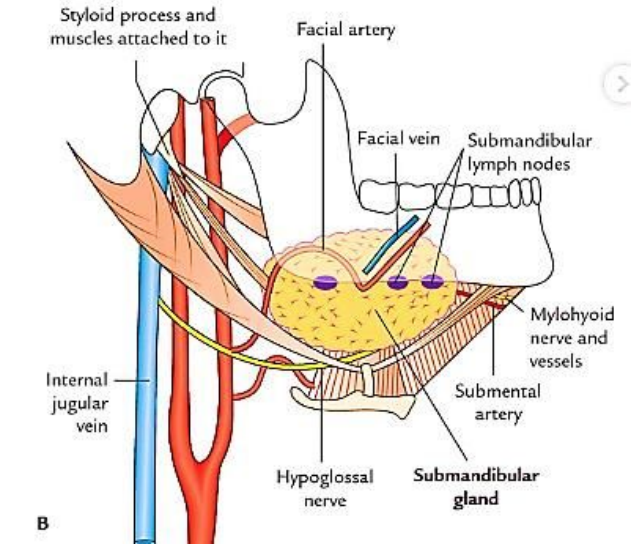
Marginal mandibular nerve and the submandibular gland

ENT Steps Anatomy, p. 190



Deep vasculature of the parotid region: arteries and veins

Gray's Atlas of Anatomy, 2nd ed., p. 507



Submandibular triangle: facial vessels and hypoglossal nerve

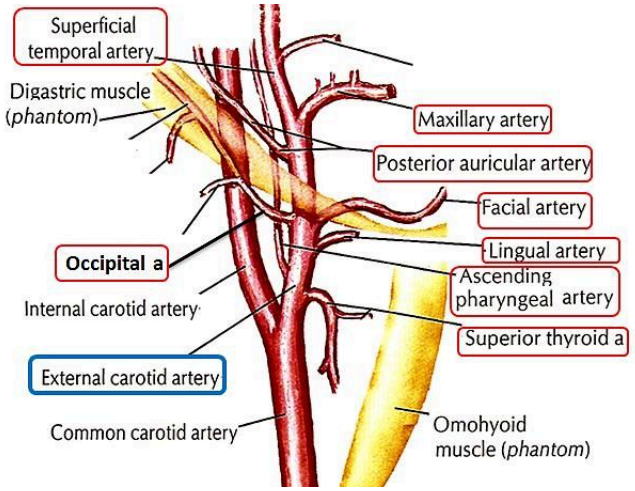
ENT Steps Anatomy, p. 271

13 Occipital, posterior auricular, ascending pharyngeal and superficial temporal arteries

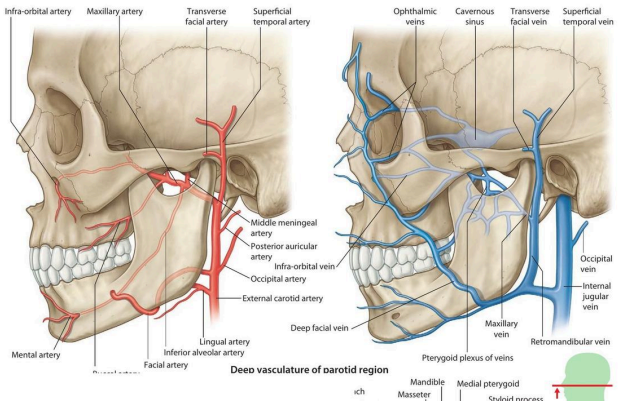
- **Ascending pharyngeal artery:** the smallest branch, from the deep (medial) side of the ECA near its origin. It climbs between the ICA and the pharynx to the skull base.
- **Its branches** go to the pharynx, the prevertebral muscles and the meninges (through the jugular foramen and hypoglossal canal), plus the **inferior tympanic artery**, which runs with Jacobson's nerve into the middle ear.
- **Occipital artery:** arises from the back of the ECA opposite the facial artery and runs back along the lower border of the **posterior belly of digastric**. The hypoglossal nerve hooks round its sternocleidomastoid branch.

- It then grooves the mastoid, medial to the digastric notch, and pierces the fascia between trapezius and SCM to supply the back of the scalp.
- **Posterior auricular artery:** arises above the digastric and runs up between the mastoid and the auricle. Its **stylomastoid branch** enters the stylomastoid foramen to supply the facial nerve, the tympanic cavity and the mastoid cells. It supplies postauricular skin flaps.
- **Superficial temporal artery:** the smaller terminal branch, beginning in the parotid behind the neck of the mandible. It crosses the root of the zygoma in front of the tragus, with the **auriculotemporal nerve just behind it**.
- **Superficial temporal branches:** transverse facial, middle temporal, anterior auricular, and frontal and parietal branches. Its pulse is felt in front of the tragus. It is the vessel taken in temporal artery biopsy and the pedicle of the temporoparietal fascia flap.

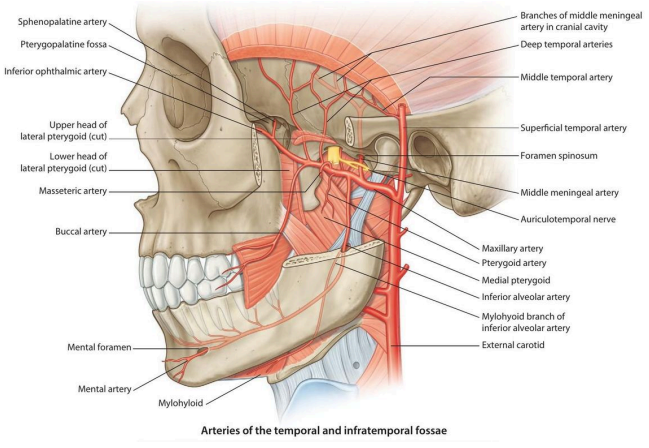
CLINICAL PEARL Middle ear paragangliomas (glomus tympanicum and jugulare) draw their main supply from the ascending pharyngeal artery (inferior tympanic branch): the usual target for preoperative embolisation.



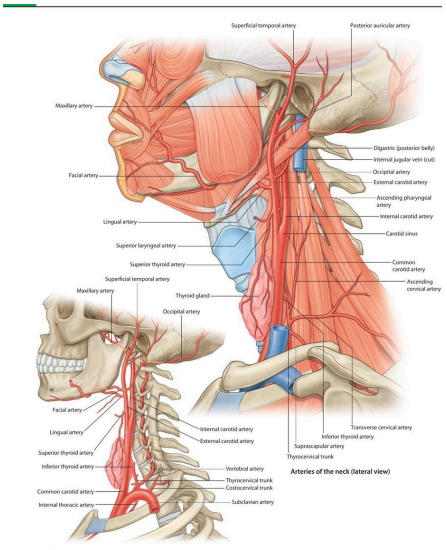
Branches of the external carotid artery
ENT Steps Anatomy, p. 267



Deep vasculature of the parotid region: arteries and veins
Gray's Atlas of Anatomy, 2nd ed., p. 507



Arteries of the temporal and infratemporal fossae: maxillary artery
Gray's Atlas of Anatomy, 2nd ed., p. 533

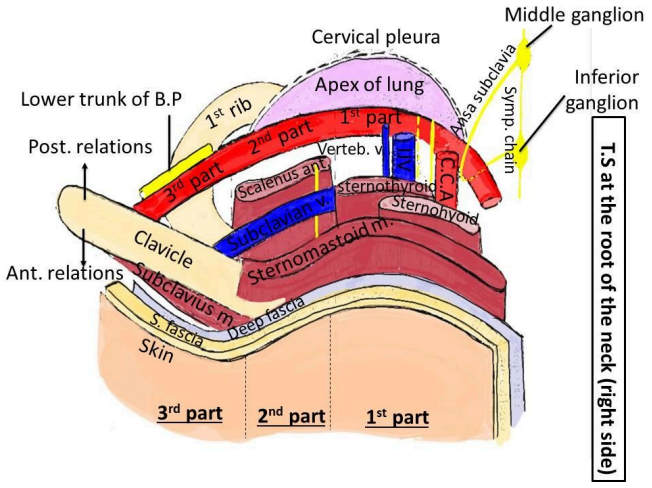


Arteries of the neck (lateral view)
Gray's Atlas of Anatomy, 2nd ed., p. 546

14 Vertebral artery and the subclavian branches

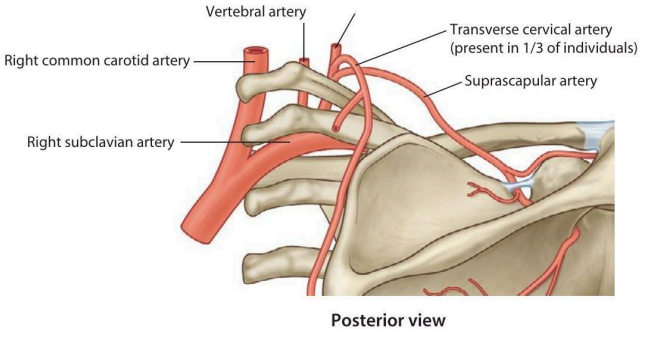
- **Vertebral artery:** the first branch of the first part of the subclavian. It has four parts (V1–V4).
- **V1 (pre-foraminal):** ascends between longus colli and scalenus anterior, at the apex of the scalenovertebral (Chassaignac's) triangle, to enter the transverse foramen of **C6** in most people.
- **V2 (foraminal):** climbs through the transverse foramina of C6 to C2, with the vertebral venous plexus and sympathetic fibres, in front of the cervical nerve roots.
- **V3 (atlantic):** turns laterally through the C1 foramen, then curves back and medially in the groove on the **posterior arch of the atlas** in the suboccipital triangle, and pierces the posterior atlanto-occipital membrane.
- **V4 (intradural):** passes through the **foramen magnum** in front of the medulla, gives the **posterior inferior cerebellar artery (PICA)** and the spinal arteries, and joins its fellow at the lower pons to form the **basilar artery**.
- **Basilar artery:** gives the **anterior inferior cerebellar artery (AICA)**, from which the **labyrinthine (internal auditory) artery** usually arises. The labyrinthine artery is an end artery to the inner ear. The basilar then gives the superior cerebellar and the posterior cerebral arteries.
- **Thyrocervical trunk** (first part of the subclavian): **inferior thyroid** (arches medially behind the carotid sheath to the back of the thyroid, crossing the RLN), **transverse cervical** and **suprascapular** arteries, plus the ascending cervical artery near the phrenic nerve.
- **Costocervical trunk** (second part, behind scalenus anterior): deep cervical and supreme intercostal arteries. The **internal thoracic** artery (first part) runs down behind the costal cartilages and is a common free-flap recipient vessel.

CLINICAL PEARL Sudden unilateral deafness with vertigo plus cerebellar or brainstem signs suggests an AICA territory infarct (the labyrinthine artery): do not label it simply as idiopathic sudden hearing loss.



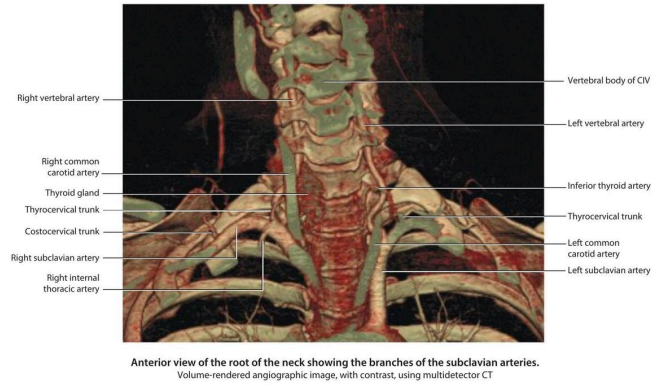
Subclavian artery and its three parts

Dr. Sameh Doss handout, p. 131



Root of the neck: thyrocervical trunk (posterior view)

Gray's Atlas of Anatomy, 2nd ed., p. 548



Branches of the subclavian arteries: CT angiography

Gray's Atlas of Anatomy, 2nd ed., p. 549

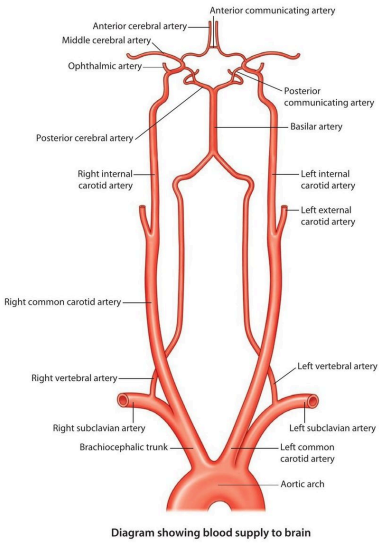
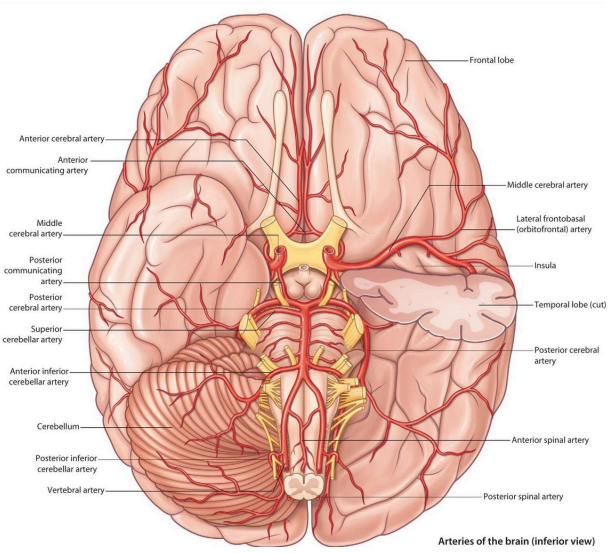


Diagram of the blood supply to the brain: carotid and vertebral systems

Gray's Atlas of Anatomy, 2nd ed., p. 500



Arteries of the brain (inferior view): vertebral, basilar and their branches

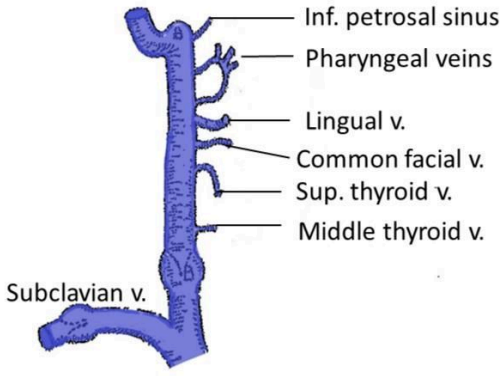
Gray's Atlas of Anatomy, 2nd ed., p. 502

15 External and anterior jugular veins

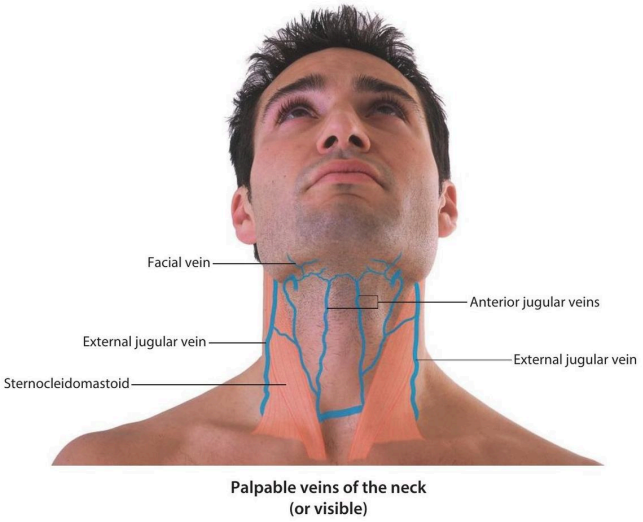
- **Retromandibular vein:** formed in the parotid by the superficial temporal and maxillary veins. It runs **deep to the facial nerve**, a landmark at parotidectomy, and splits into an anterior division (to the facial vein) and a posterior division.
- **External jugular vein:** formed near the angle of the mandible by the **posterior division of the retromandibular vein** and the **posterior auricular vein**. It runs from the angle of the mandible towards the middle of the clavicle, crossing SCM obliquely under platysma, alongside the great auricular nerve.
- **Termination:** it pierces the deep fascia of the posterior triangle about 2–3 cm above the clavicle and drains into the **subclavian vein**. Its tributaries are the posterior external jugular, transverse cervical, suprascapular and anterior jugular veins.
- Where it pierces the fascia the vein wall is held open: a cut vein here can suck in air (air embolism) and should be clamped at once.

- **Anterior jugular veins:** begin near the hyoid from small submental veins and run down beside the midline on the strap muscles.
- Above the sternum they are linked by the **jugular venous arch** in the suprasternal space (of Burns), then turn laterally deep to SCM to join the external jugular vein. The arch is met in tracheostomy and thyroid surgery.
- **Brachiocephalic veins:** each IJV joins the subclavian vein behind the sternoclavicular joint. The **left brachiocephalic vein** crosses the superior mediastinum behind the manubrium and can rise into the neck in children, where it is at risk at low tracheostomy.

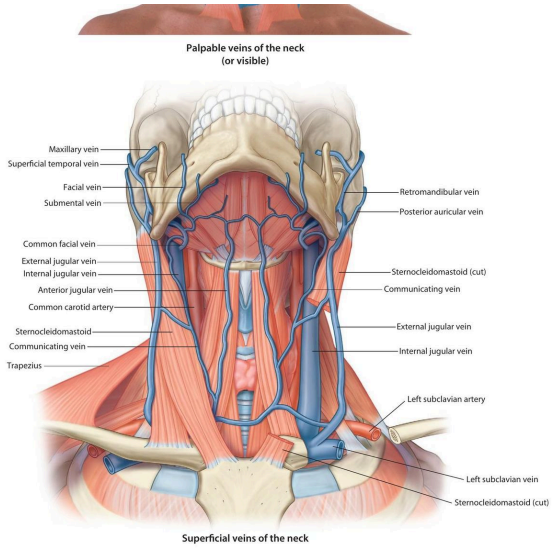
> **CLINICAL PEARL** Keep low tracheostomy incisions above the sternal notch and dissect in the midline: the jugular arch and, in children, a high left brachiocephalic vein lie just below.



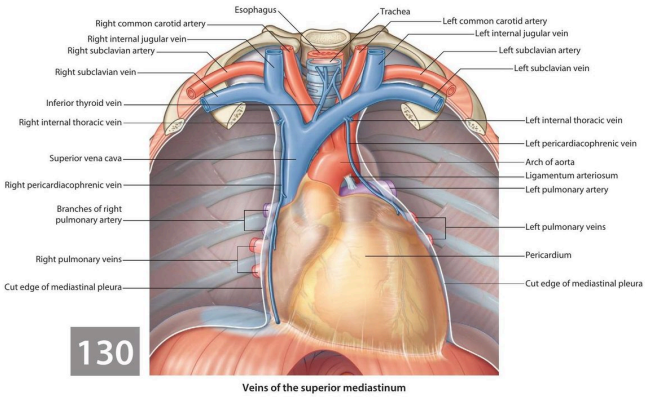
Tributaries of the internal jugular vein
Dr. Sameh Doss handout, p. 146



Palpable veins of the neck
Gray's Atlas of Anatomy, 2nd ed., p. 539



Superficial veins of the neck
Gray's Atlas of Anatomy, 2nd ed., p. 539



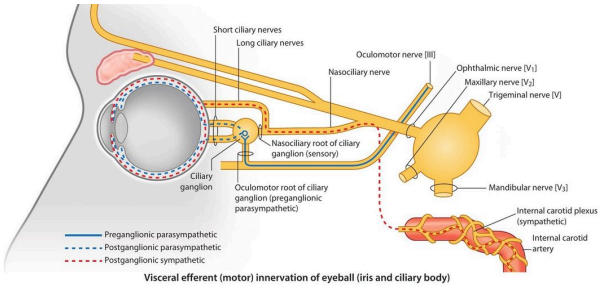
Veins of the superior mediastinum: brachiocephalic veins and their tributaries
Gray's Atlas of Anatomy, 2nd ed., p. 109

16 Horner's syndrome: the oculosympathetic pathway

- **First-order neuron:** hypothalamus → down through the lateral brainstem and cervical cord → synapse in the **ciliospinal centre of Budge (C8–T2)**.

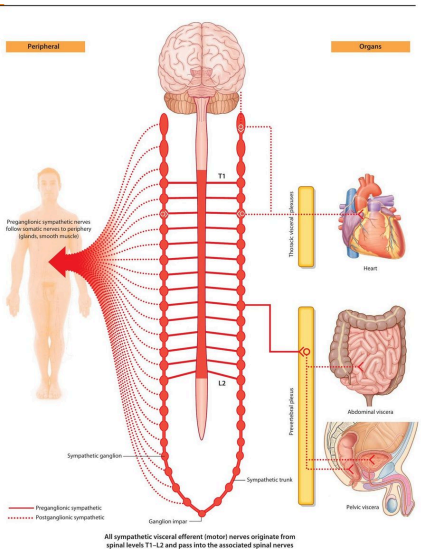
- **Second-order (preganglionic) neuron:** leaves in the **T1 root**, arches over the **apex of the lung**, and ascends in the cervical sympathetic chain behind the carotid sheath to synapse in the **superior cervical ganglion (C2–C3)**.
- **Third-order (postganglionic) neuron:** runs on the **internal carotid artery** through the carotid canal and the cavernous sinus, joins V1 and the nasociliary nerve, and reaches the **dilator pupillae** via the long ciliary nerves and **Müller's (superior tarsal) muscle**.
- **Sweat fibres** to most of the face leave the superior cervical ganglion with the **external carotid** branches, so third-order lesions largely spare facial sweating.
- **Signs:** mild ptosis with a raised lower lid (apparent enophthalmos), **miosis** with dilation lag in dim light, and anhidrosis whose extent shows the level. Iris heterochromia is seen if the syndrome is congenital.
- **ENT causes:** neck and thyroid surgery, neck dissection, **carotid dissection** (painful acute Horner's), Pancoast tumour, parapharyngeal and skull-base tumours, nasopharyngeal carcinoma, cervical sympathetic schwannoma.
- **Drop tests:** apraclonidine reverses the anisocoria (denervation supersensitivity), and cocaine fails to dilate the Horner pupil. Hydroxyamphetamine dilates it in first- and second-order lesions but not in third-order ones.

> **CLINICAL PEARL** A new painful Horner's syndrome with neck or facial pain is a carotid dissection until imaging proves otherwise.



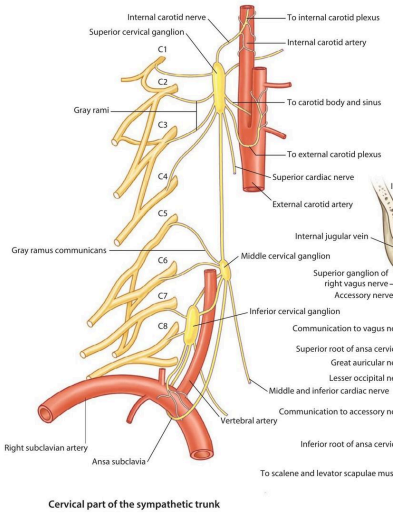
Autonomic innervation of the eyeball: sympathetic and parasympathetic pathways to the iris

Gray's Atlas of Anatomy, 2nd ed., p. 517



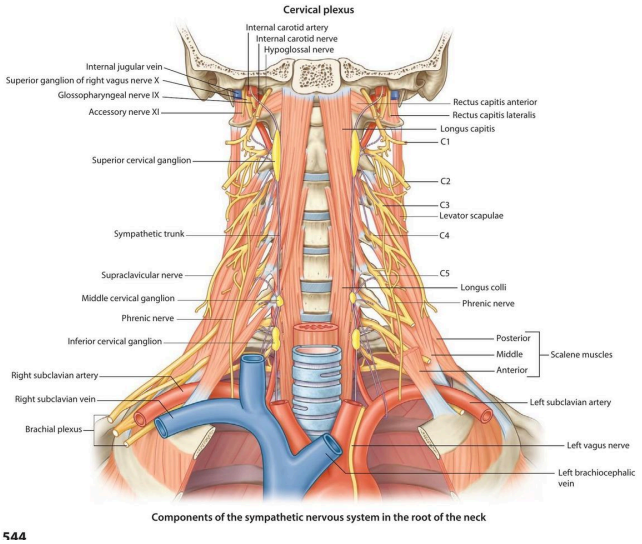
Sympathetic outflow: T1 to L2, the sympathetic trunk and its targets

Gray's Atlas of Anatomy, 2nd ed., p. 14



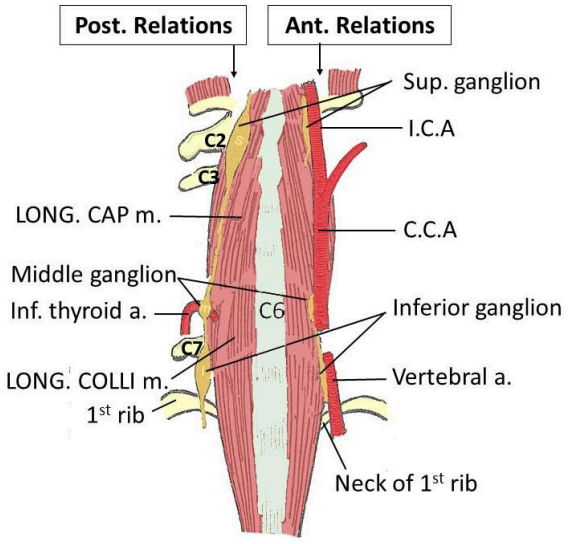
Cervical part of the sympathetic trunk

Gray's Atlas of Anatomy, 2nd ed., p. 545



Sympathetic system in the root of the neck

Gray's Atlas of Anatomy, 2nd ed., p. 544



Cervical sympathetic chain

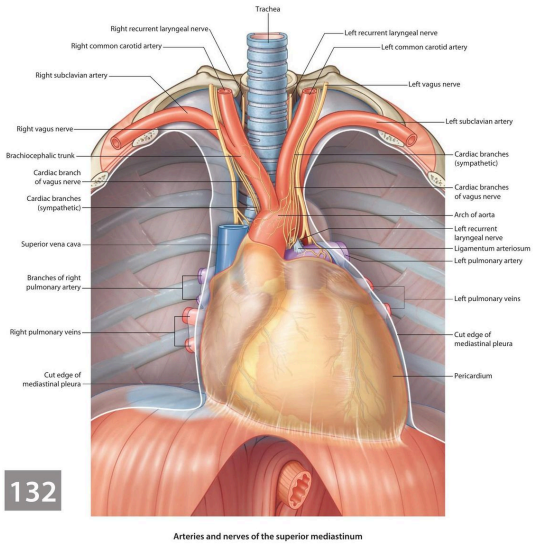
Dr. Sameh Doss handout, p. 160

17 Variations surgeons meet

- **Non-recurrent laryngeal nerve:** almost always on the **right** (well under 1% of people), with an **aberrant right subclavian artery** (arteria lusoria) arising last from the arch and passing behind the oesophagus.
- The non-recurrent nerve leaves the vagus high in the neck and runs straight to the larynx, often near the inferior thyroid artery. A retro-oesophageal subclavian on preoperative CT predicts it. A left non-recurrent nerve needs a right-sided arch with situs inversus and is exceptionally rare.
- **RLN and the inferior thyroid artery:** the nerve may lie in front of, behind or between the artery's branches, and the two sides often differ. **Extralaryngeal branching** before the nerve enters the larynx is common; the motor fibres usually run in the anterior branch.
- **Tubercle of Zuckerkandl:** this posterior extension of the thyroid usually lies **superficial (lateral)** to the RLN, so the nerve is found deep to it.
- **Carotid bifurcation:** usually at the upper border of the thyroid cartilage (C3–C4), but it can be as high as the angle of the mandible. The ECA occasionally lies lateral to the ICA at its origin.

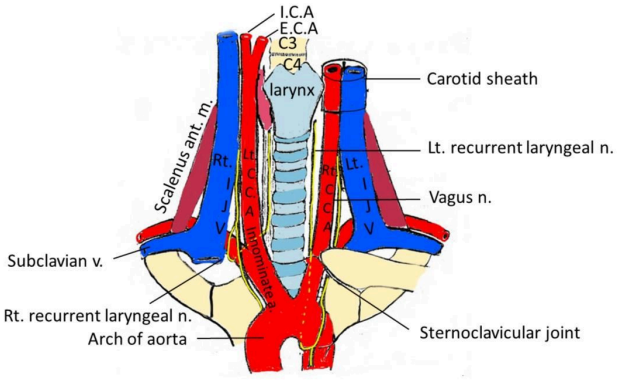
- **Medialised (tortuous) ICA:** the ICA can loop medially behind the posterior pharyngeal wall or tonsil, sometimes reaching the midline. A pulsating pharyngeal wall is a warning before adenoid, tonsil or pharyngeal surgery.
- **Common trunks:** linguofacial and thyrolingual trunks from the ECA are common.
- **Accessory nerve:** usually lateral (superficial) to the IJV near the atlas; medial in a minority, and occasionally through the vein.
- **Marginal mandibular nerve:** can dip below the lower border of the mandible, behind the facial vessels, especially with the neck extended.
- **Vertebral artery:** occasionally arises from the arch directly (more often on the left) and may enter a higher transverse foramen than C6.
- **Thoracic duct:** usually ends at the left jugulosubclavian junction, but can be double or end on the right.

> **CLINICAL PEARL** Variations are why surgeons identify the nerve or vessel itself rather than trusting a single landmark.



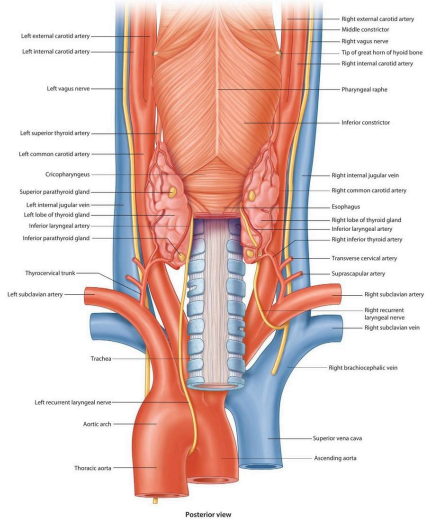
Arteries and nerves of the superior mediastinum: vagus and recurrent laryngeal loops

Gray's Atlas of Anatomy, 2nd ed., p. 110



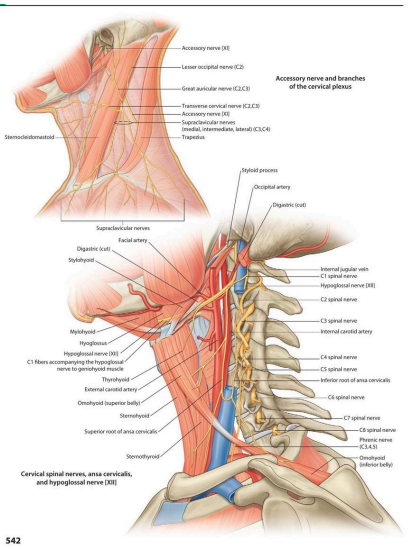
Root of the neck: carotids, subclavian and recurrent laryngeal nerves

Dr. Sameh Doss handout, p. 135



Vasculature of the thyroid gland (posterior view): parathyroids and recurrent laryngeal nerves

Gray's Atlas of Anatomy, 2nd ed., p. 565



Accessory nerve, cervical plexus, ansa cervicalis and hypoglossal nerve

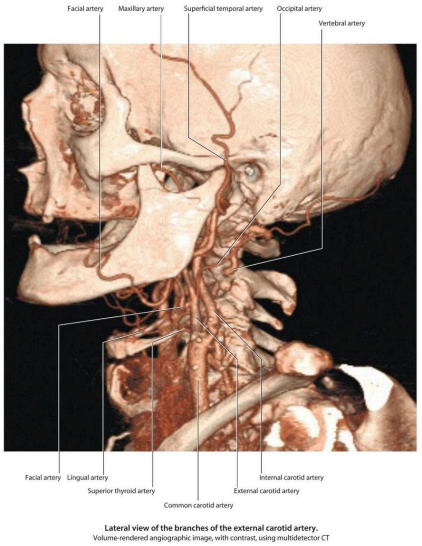
Gray's Atlas of Anatomy, 2nd ed., p. 542

18 Surgical anatomy: In theatre

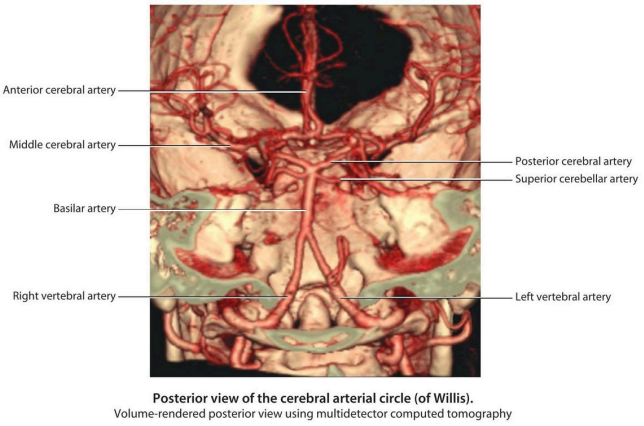
- **ECA ligation:** confirm branches before tying; tie above the superior thyroid artery.
- **Carotid body tumour:** Shamblin I (small, localised, minimal vessel attachment), II (partly surrounds the vessels), III (encases them); risk to X and XII rises with grade.
- **IJV ligation:** one side is safe; bilateral simultaneous ligation risks facial and cerebral oedema, so stage it.
- **Accessory nerve** at level II: usually lateral (superficial) to the IJV, occasionally medial or through it.

19 Radiological anatomy: On imaging

- **CT/MR angiography:** carotid body tumours splay the ICA and ECA (lyre sign); glomus vagale pushes both forward without splaying.
- **Doppler ultrasound** for carotid disease and IJV thrombosis (Lemierre's syndrome after pharyngitis).



Branches of the external carotid artery: CT angiography
Gray's Atlas of Anatomy, 2nd ed., p. 547



Cerebral arterial circle: CT angiography, posterior view
Gray's Atlas of Anatomy, 2nd ed., p. 501

T Tables

Branches of the external carotid artery

Branch	Main territory
Superior thyroid	Thyroid upper pole, larynx (superior laryngeal), SCM
Ascending pharyngeal	Pharynx, tonsil, middle ear, meninges
Lingual	Tongue, floor of mouth, tonsil (dorsal lingual)
Facial	Face, tonsil (tonsillar branch), submandibular gland, lips
Occipital	Back of scalp, SCM
Posterior auricular	Auricle, scalp, stylomastoid artery to middle ear
Maxillary	Deep face, nose (sphenopalatine), middle meningeal, teeth, middle ear
Superficial temporal	Temporal scalp, parotid

The last four cranial nerves

Nerve	Exit	Motor	Sensory / other	Lesion sign
IX	Jugular foramen	Stylopharyngeus	Posterior third of tongue, tonsil, middle ear, carotid sinus	Lost gag reflex (afferent)
X	Jugular foramen	Pharynx, palate (except tensor), larynx	Larynx, ear (Arnold's), viscera	Uvula deviates to healthy side; cord palsy
XI	Jugular foramen	SCM, trapezius	—	Shoulder drop, weak head turning
XII	Hypoglossal canal	Tongue muscles (except palatoglossus)	—	Tongue deviates to weak side

M Mnemonics

External carotid branches

Some Anatomists Like Freaking Out Poor Medical Students

Superior thyroid, Ascending pharyngeal, Lingual, Facial, Occipital, Posterior auricular, Maxillary, Superficial temporal

Phrenic nerve

C3, 4, 5 keeps the diaphragm alive

Roots of the phrenic nerve

Carotid sheath contents

I See 10 CC's in the IV

Internal jugular vein, vagus (X), Common/internal Carotid

Subclavian first-part branches

VIT

Vertebral, Internal thoracic, Thyrocervical trunk

Thyrocervical trunk

STAIn

Suprascapular, Transverse cervical, Inferior thyroid (which gives the Ascending cervical)

Horner's

PAM

Ptosis, Anhidrosis, Miosis

Q Rapid recall

1. **Level of the carotid bifurcation?** Upper border of the thyroid cartilage (C3–C4).
2. **Branches of the external carotid artery in order?** Superior thyroid, ascending pharyngeal, lingual, facial, occipital, posterior auricular, maxillary, superficial temporal.
3. **How do you tell the ICA from the ECA in the neck?** The ICA has no branches in the neck; the ECA does.
4. **Nerve supply of the carotid sinus and body?** Glossopharyngeal nerve (nerve of Hering).
5. **Relation of the lingual artery and hypoglossal nerve to hyoglossus?** Artery deep to hyoglossus; nerve on its surface.
6. **Roots of the ansa cervicalis?** Superior root C1 (travelling with XII), inferior root C2–C3.
7. **Roots and course of the phrenic nerve in the neck?** C3–C5, on scalenus anterior deep to the prevertebral fascia.
8. **Where does the thoracic duct end?** Junction of the left IJV and subclavian vein.
9. **Branches of the first part of the subclavian artery?** Vertebral, internal thoracic, thyrocervical trunk.
10. **Branches of the thyrocervical trunk?** Inferior thyroid, transverse cervical, suprascapular.
11. **Where does the ECA end?** In the parotid behind the neck of mandible, dividing into maxillary and superficial temporal arteries.
12. **Shamblin classification?** I small and localised with minimal vessel attachment, II partly surrounds, III encases the carotids.

13. **Origin of the IJV?** Continuation of the sigmoid sinus at the jugular foramen.
14. **Which nerve gives the carotid sinus branch?** Glossopharyngeal (IX).
15. **What is Lemierre's syndrome?** IJV thrombophlebitis after pharyngitis, usually Fusobacterium necrophorum, with septic emboli.
16. **Ganglia of the cervical sympathetic chain?** Superior (C2–3), middle (C6), inferior (often fused with T1 as the stellate ganglion).
17. **Signs of Horner's syndrome?** Ptosis, miosis, anhidrosis, apparent enophthalmos.
18. **What is Chassaignac's tubercle?** The anterior tubercle of C6, apex of the scalenovertebral triangle.
19. **Which is usually the first branch of the external carotid artery?** The superior thyroid artery, just below the greater horn of the hyoid.
20. **What runs with the superior laryngeal artery through the thyrohyoid membrane?** The internal laryngeal nerve.
21. **Lingual artery and hypoglossal nerve: which is deep to hyoglossus?** The lingual artery runs deep to hyoglossus; the hypoglossal nerve runs on its surface.
22. **Boundaries of Lesser's triangle?** The anterior and posterior bellies of digastric and the hypoglossal nerve; the lingual artery lies deep to its floor (hyoglossus).
23. **Where does the facial vein lie relative to the submandibular gland?** Superficial to it; the facial artery is deep to the gland.
24. **Two venous routes from the face to the cavernous sinus?** Angular vein to the superior ophthalmic vein; deep facial vein to the pterygoid plexus.
25. **What is the Hayes Martin manoeuvre?** Dividing the facial vein low and turning it up with the flap to protect the marginal mandibular nerve.
26. **Which artery is the main supply of middle ear paragangliomas?** The ascending pharyngeal artery, through its inferior tympanic branch.
27. **Which branch of the posterior auricular artery supplies the facial nerve in its canal?** The stylomastoid artery.
28. **What lies just behind the superficial temporal artery in front of the tragus?** The auriculotemporal nerve.
29. **Which transverse foramen does the vertebral artery usually enter?** C6.
30. **Where does the labyrinthine artery usually arise?** From the anterior inferior cerebellar artery (AICA), a branch of the basilar.
31. **Branches of the thyrocervical trunk?** Inferior thyroid, transverse cervical and suprascapular (and ascending cervical from the inferior thyroid).
32. **How does the external jugular vein end?** It pierces the deep fascia 2–3 cm above the clavicle and drains into the subclavian vein.
33. **Why is a cut external jugular vein low in the neck dangerous?** Its wall is held open by the fascia it pierces, so air can be sucked in (air embolism).
34. **Three neurons of the oculosympathetic pathway?** Hypothalamus to C8–T2; C8–T2 (via T1 and the lung apex) to the superior cervical ganglion; superior cervical ganglion along the ICA to the eye.
35. **Why does a postganglionic Horner's spare facial sweating?** Facial sudomotor fibres leave the superior cervical ganglion with the external carotid branches, not the ICA.
36. **Which drop test separates third-order Horner's from the others?** Hydroxyamphetamine: it fails to dilate the pupil only when the third-order neuron is damaged.
37. **With which vascular anomaly does a non-recurrent right laryngeal nerve occur?** An aberrant right subclavian artery (arteria lusoria) passing behind the oesophagus.
38. **Where does the RLN lie relative to the tubercle of Zuckerkandl?** Usually deep (medial) to it.
39. **High vagal lesion: which way does the uvula deviate?** Towards the healthy side.
40. **Accessory nerve injured in the posterior triangle: which muscle is spared?** Sternocleidomastoid (its branch leaves above the triangle).
41. **Hypoglossal lower motor neuron lesion: which way does the tongue deviate?** Towards the side of the lesion, with wasting and fasciculation.

Figures from ENT Steps Anatomy (based on Bansal) and Dr. Sameh Doss's Anatomy of Head & Neck handout, for personal study.